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Oefeningenreeks 5J nr. 2.4

$$\begin{cases} x(t) = a\cos^3(t) \\ y(t) = a\sin^3(t) \end{cases}$$

$$x'(t) = -3a\cos^2(t)\sin(t)$$

$$y'(t) = 3a\sin^2(t)\cos(t)$$

$$S = \int_0^{2\pi} \sqrt{9a^2\cos^4(t)\sin^2(t) + 9a^2\cos^2(t)\sin^4(t)dt}$$

$$= 3a\int_0^{2\pi} \sqrt{\cos^2(t)\sin^2(t)(\cos^2(t) + sin^2(t))}dt$$

$$= 3a\int_0^{2\pi} \cos(t)\sin(t)dt$$

$$= 4 \cdot 3a\int_0^{\frac{\pi}{2}} \cos(t)\sin(t)dt$$

Stel $x = \sin(t) \Rightarrow dx = \cos(t)dt$

$$= \frac{4 \cdot 3}{2}a\left[\sin^2(t)\right]_0^{\frac{\pi}{2}}$$

$$= 6a$$

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References